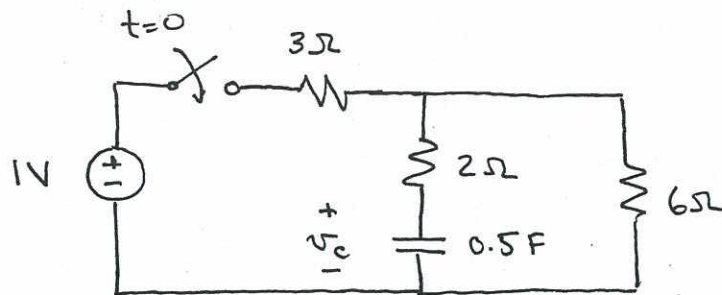
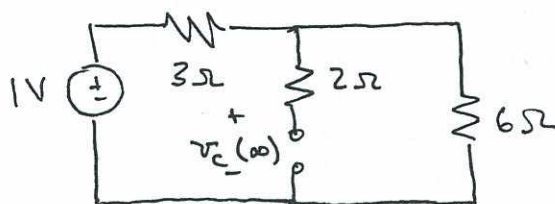


7.1



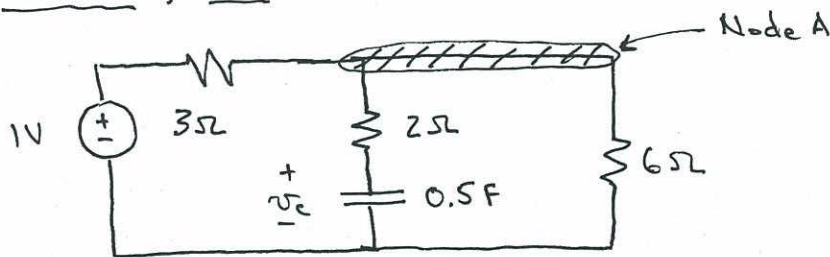
Initial condition : $v_c(t=0^+) = v_c(t=0^-) = \underline{\underline{0V}} = v_c(0)$

Final condition :



$$v_c(t \rightarrow \infty) = \left(\frac{6\Omega}{9\Omega} \right) 1V = \underline{\underline{\frac{2}{3}V}}$$

Circuit, $t > 0$:



Nodal analysis (KCL at A):

$$\frac{1}{3\Omega} \left[1 - (v_c + 2(0.5F) \frac{dv_c}{dt}) \right] = (0.5F) \frac{dv_c}{dt} + \left[v_c + 2(0.5F) \frac{dv_c}{dt} \right] \frac{1}{6\Omega}$$

$$2 - 2v_c - 2 \frac{dv_c}{dt} = 3 \frac{dv_c}{dt} + v_c + \frac{dv_c}{dt}$$

$$2 = 6 \frac{dv_c}{dt} + 3v_c$$

$$\underline{\underline{\frac{dv_c}{dt} + \frac{1}{2} v_c = \frac{1}{3}}}$$

Differential equation governing $v_c(t)$, $t > 0$

↓ (continued)

7.1 (cont'd)

Governing differential equation: $\frac{dv_c}{dt} + \frac{1}{2} v_c = \frac{1}{3}$

$$\tau = 2 \text{ seconds}$$

Form of solution:

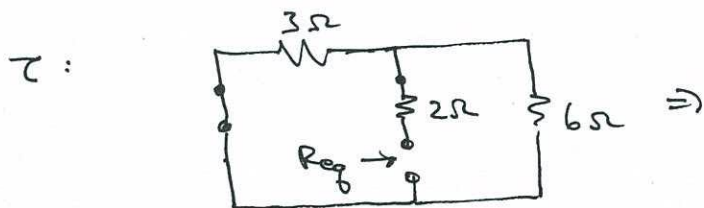
$$v_c(t) = K_1 + K_2 e^{-t/\tau}$$

Initial, final conditions:

$$\left. \begin{array}{l} v_c(0) = 0 = K_1 + K_2 \\ v_c(\infty) = \frac{2}{3} = K_1 \end{array} \right\} \Rightarrow K_1 = \frac{2}{3}, K_2 = -\frac{2}{3}$$

$$v_c(t) = \frac{2}{3} [1 - e^{-t/2}] \text{ V}, t > 0$$

check τ , steady-state response:



$$R_{eq} = 2\Omega + \frac{(6\Omega)(3\Omega)}{6\Omega + 3\Omega} = 4\Omega$$

$$\tau = R_{eq}C = (4)(0.5) = 2 \text{ sec} \checkmark$$

steady-state response:

$$\frac{dv_c(\infty)}{dt} + \frac{1}{2} v_c(\infty) = \frac{1}{3} \Rightarrow v_c(\infty) = \frac{2}{3} \text{ V} \quad \left(\begin{array}{l} \text{matches} \\ \text{final} \\ \text{condition} \end{array} \right)$$

continued

7.1 (cont'd)

(b) capacitance = 1F

changing capacitance only changes the time constant. Equivalent resistance is same as in part (a), so:

$$\tau = R_{eq}C = (4\Omega)(1F) = 4\text{ sec}$$

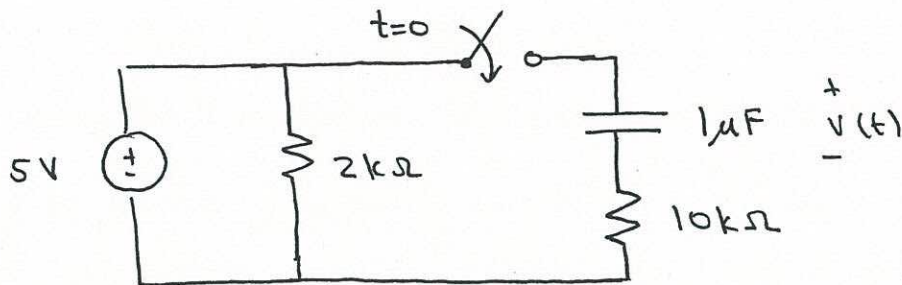
$$v_c(t) = \frac{2}{3} [1 - e^{-t/4}], \quad t > 0$$

(c) capacitance = 0.25F

Same approach as (b):

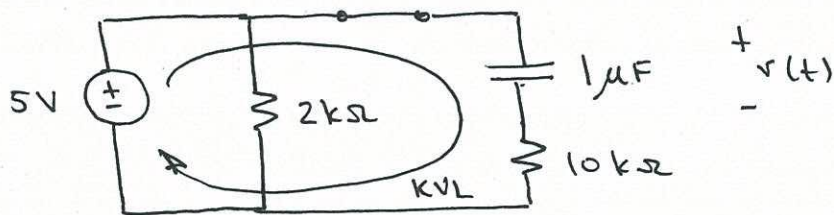
$$v_c(t) = \frac{2}{3} [1 - e^{-t}], \quad t > 0$$

7.2



Initial condition: $v(t=0^+) = v(t=0^-) = \underline{\underline{0V}}$

Circuit, for $t > 0$:

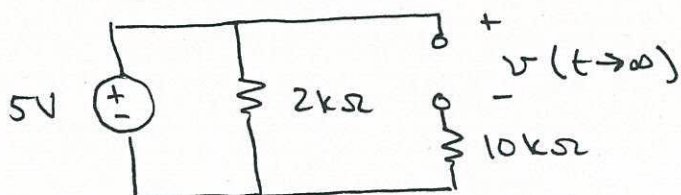


$$\text{KVL: } 5V = v(t) + (10k\Omega) \left[1\mu F \frac{dv(t)}{dt} \right]$$

$$5V = v + 10^{-2} \frac{dv}{dt} \Rightarrow \frac{dv}{dt} + 100v = 500$$

$$\tau = \underline{\underline{\frac{1}{100} \text{ sec}}}$$

Final condition:



$$\Rightarrow \underline{\underline{v(t \rightarrow \infty) = 5V}}$$

Form of solution:

$$v(t) = K_1 + K_2 e^{-t/\tau}$$

$$\tau = \frac{1}{100}$$

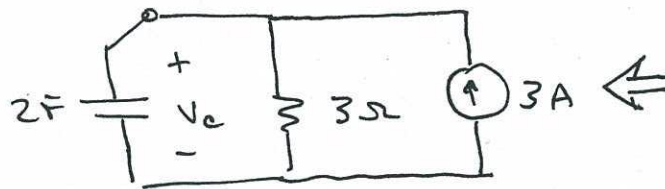
$$v(0) = 0, v(\infty) = 5$$

$$\boxed{v(t) = 5(1 - e^{-100t}), t > 0}$$

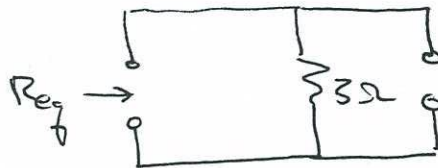
7.3

- Solution is of form: $v(t) = K_1 + K_2 e^{-t/\tau}$
- Find τ from RC time constant

Circuit for $t > 0$:



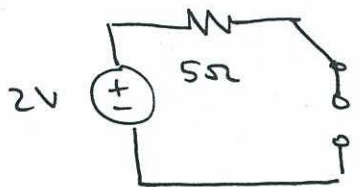
Kill sources, find R_{eq} :



$$\Rightarrow R_{eq} = 3\Omega \Rightarrow R_{eq}C = \underline{\underline{6 \text{ sec}}}$$

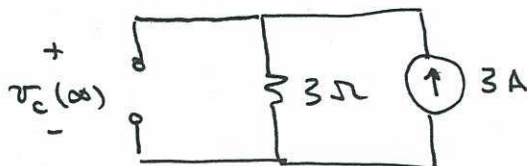
$$\tau = \underline{\underline{6 \text{ sec.}}}$$

- Initial condition (circuit at $t < 0$):



$$v_c(t=0^-) = \underline{\underline{2V}} \Rightarrow \underline{\underline{2V = K_1 + K_2}}$$

- Final Condition, (circuit as $t \rightarrow \infty$):



$$v_c(t \rightarrow \infty) = (3\Omega)(3A) = \underline{\underline{9V}}$$

$$\underline{\underline{9V = K_1}}$$

$$K_1 = 9V$$

$$K_2 = -7V$$

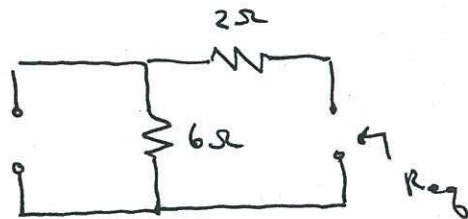
$$\tau = 6 \text{ sec}$$

$$\boxed{v(t) = 9 - 7e^{-t/6} \text{ V}}$$

7.4

Solution is of the form $y(t) = K_1 + K_2 e^{-t/\tau}$.

Finding τ from RC:



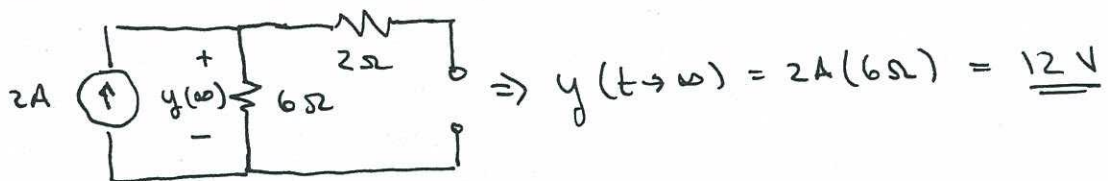
$$\Rightarrow R_{eq} = 2\Omega + 6\Omega = 8\Omega$$

$$\tau = (8\Omega)(\frac{1}{4}F) = \underline{2\text{sec}}$$

Initial condition

$$y(t=0) = 0V \quad (\text{no current applied before } t=0)$$

Final condition:



Solving for K_1 & K_2 :

$$\left. \begin{array}{l} y(0) = K_1 + K_2 = 0V \\ y(\infty) = K_1 = 12V \end{array} \right\} \Rightarrow \begin{array}{l} K_1 = 12V \\ K_2 = -12V \end{array}$$

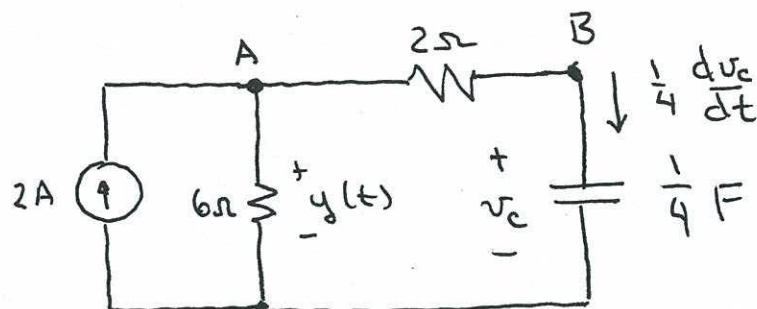
$$y(t) = 12(1 - e^{-t/2})V$$

Let's check this result by writing the diff. eqn. governing the circuit

} continued

7.4 (cont'd)

circuit for $t > 0$:



KVL around right-most loop:

$$y(t) = (2\Omega) \left(\frac{1}{4} \frac{dv_c}{dt} \right) + v_c \Rightarrow \underline{\frac{1}{2} \frac{dv_c}{dt} + v_c = y} \quad (*)$$

KCL at A:

$$2A = \frac{y}{6\Omega} + \frac{1}{4} \frac{dv_c}{dt}$$

This equation is fine, but looks like it will get messy when combined with the above. Let's try something else

Re-do KCL at A:

$$2A = \frac{y}{6\Omega} + \frac{y - v_c}{2\Omega}$$

$$12 = y + 3(y - v_c) \Rightarrow v_c = \frac{4}{3}y - 4 \Rightarrow \underline{\frac{dv_c}{dt} = \frac{4}{3} \frac{dy}{dt}}$$

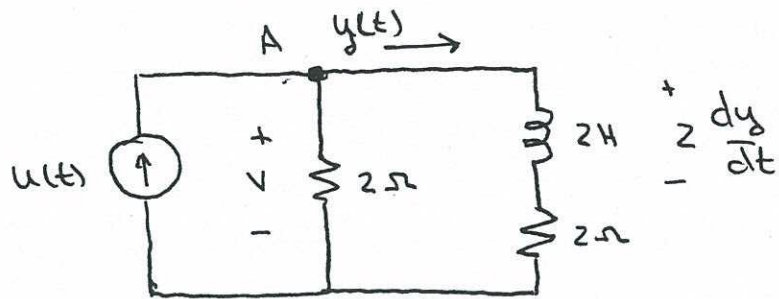
These two are easy to substitute into eqn. (*)

Diff. eqn. becomes:

$$y = \frac{1}{2} \left[\frac{4}{3} \frac{dy}{dt} \right] + \left[\frac{4}{3}y - 4 \right] \Rightarrow \underline{\frac{dy}{dt} + \frac{1}{2}y = 6} \Rightarrow y(\infty) = 12V$$

$\tau = 2 \text{ sec}$
 $\Rightarrow$ AGREES with previous solution!

7.5

(a) circuit for $t > 0$:

$$\text{KCL at A} \Rightarrow u = \frac{v}{2} + y$$

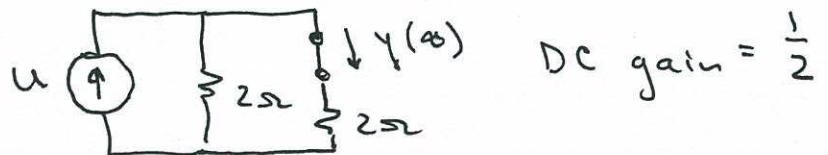
$$\text{KVL, right loop} \Rightarrow v = 2 \frac{dy}{dt} + 2y$$

$$\boxed{\frac{dy}{dt} + 2y = u}$$

- check the diff. equation based on expected circuit response.

$$\text{D.E : } \tau = \frac{1}{2}, \text{ DC gain} = \frac{Y(\omega)}{u(\omega)} = \frac{1}{2}$$

$$\text{circuit : } \tau = \frac{L}{R_{eq}} = \frac{2H}{4\Omega} = \frac{1}{2},$$



$\Rightarrow$ The Diff. eqn. and expected circuit response agree. ✓

↓ continued

7.5 (cont'd)

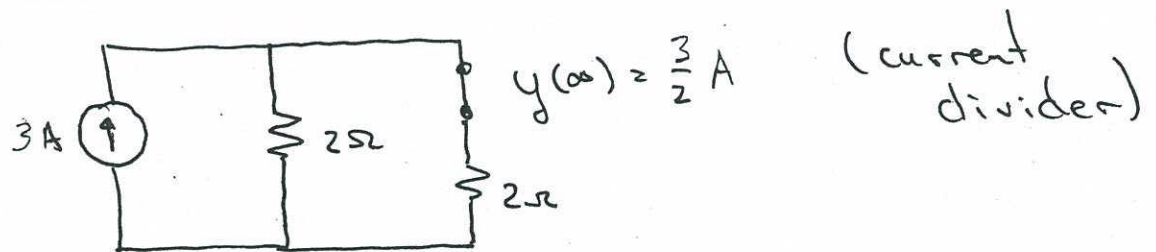
(b) From (a), $\frac{dy}{dt} + 2y = u = 3u_0(t) = \begin{cases} 3A, & t \geq 0 \\ 0A, & t < 0 \end{cases}$

form of solution (if u is a step function) is:

$$y(t) = K_1 + K_2 e^{-t/\tau}$$

$y(0) = 0A$ (no current through inductor before step is applied.)

Final condition:



Solving for K_1 & K_2

$$\left. \begin{aligned} y(0) &= K_1 + K_2 = 0 \\ y(\infty) &= K_1 = \frac{3}{2} \end{aligned} \right\} \Rightarrow K_1 = \frac{3}{2}, \quad K_2 = -\frac{3}{2}$$

$$\boxed{y(t) = \frac{3}{2} (1 - e^{-2t})}$$

(c) In part (a), we

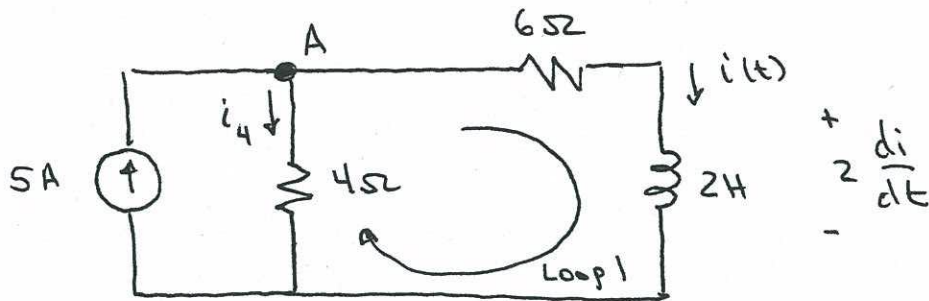
saw that the circuit's time constant

$$\tau = \frac{L}{R_{eq}} = \frac{2H}{4\Omega} = \frac{1}{2} \text{ sec. and the DC gain}$$

is $\frac{1}{2}$. Therefore, the above solution

agrees with what we expect!

7.6

(a) circuit for $t > 0$:

KCL at A: $i_4 = 5 - i(t)$ (*)

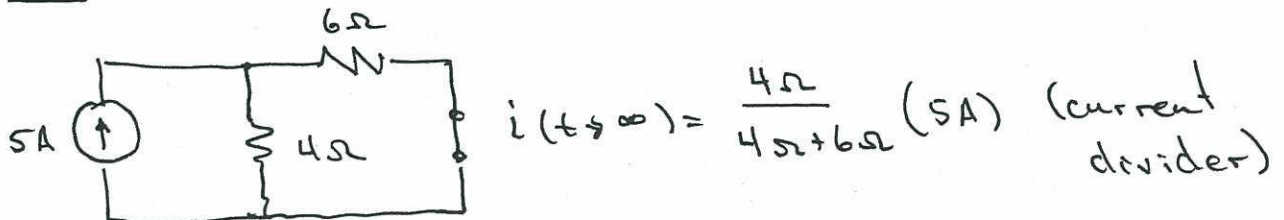
KVL, Loop 1: $(4\Omega) i_4 = (6\Omega) i + (2H) \frac{di}{dt}$ (**)

Substitute (*) into (**):

$$4(5 - i) = 6i + 2 \frac{di}{dt} \Rightarrow \boxed{20 = 10i + 2 \frac{di}{dt}}$$

(b) Initial: $i(t=0) = \underline{\underline{0A}}$ (no current before switch closes)

Final: Inductor looks like short circuit as $t \rightarrow \infty$:



$$i(t \rightarrow \infty) = \frac{4}{10} (5) = \underline{\underline{2A}}$$

↓ continued.

7.6 (cont'd)

(c) Form of solution:

$$i(t) = K_1 + K_2 e^{-t/\tau}$$

From diff. eqn.: $20 = 10i + 2 \frac{di}{dt} \Rightarrow \frac{di}{dt} + 5i = 10$

$$\tau = \frac{1}{5}$$

From initial, final conditions:

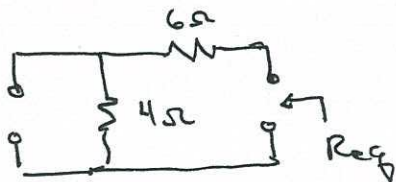
$$\left. \begin{array}{l} y(0) = K_1 + K_2 = 0 \text{ A} \\ y(\infty) = K_1 = 2 \text{ A} \end{array} \right\} \Rightarrow \begin{array}{l} \underline{K_1 = 2 \text{ A}} \\ \underline{K_2 = -2 \text{ A}} \end{array}$$

Solution is:

$$i(t) = 2(1 - e^{-5t}) \text{ A}$$

Check results based on circuit (Not required, but good!)

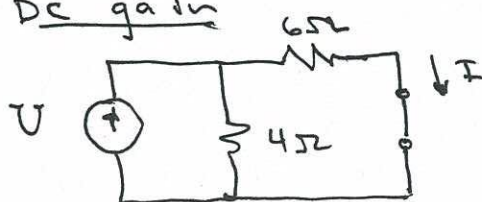
• Time constant: $\tau = \frac{L}{R_{eq}}$



$$\Rightarrow R_{eq} = 10 \Omega$$

$$\tau = \frac{2 \text{ H}}{10 \Omega} = \frac{1}{5} \text{ sec} \checkmark$$

• DC gain

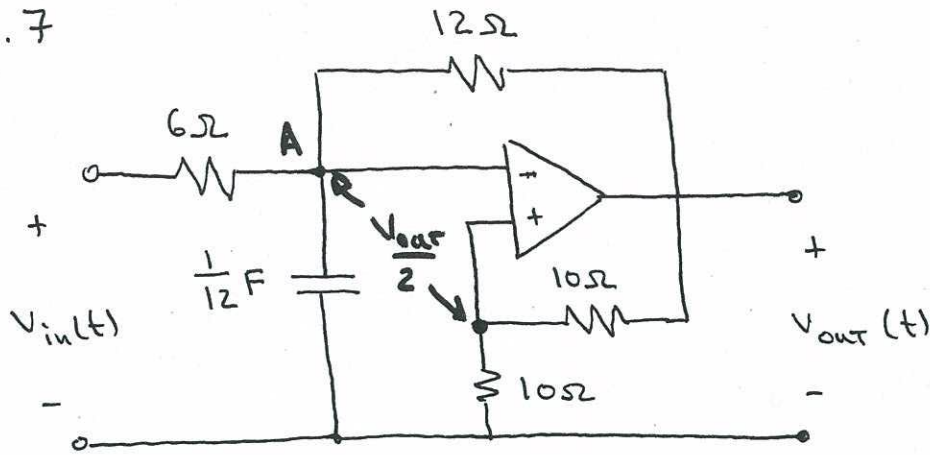


$$I = \frac{4}{6+4} U \Rightarrow \frac{I}{U} = \frac{4}{10} = \frac{2}{5}$$

$$\Rightarrow \text{Apply 5A input, } i(\infty) \rightarrow 2 \text{ A} \checkmark$$

Both consistent with above.

7.7



Voltages at both op-amp input nodes are $\frac{V_{out}}{2}$.

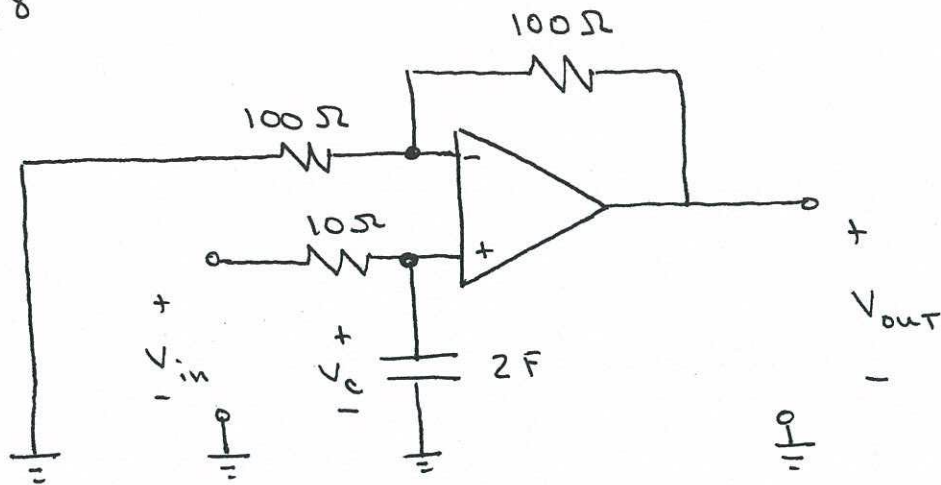
KCL at A:

$$\frac{V_{in} - \frac{V_{out}}{2}}{6\Omega} = \left(\frac{1}{12}F\right) \frac{d}{dt} \left[\frac{V_{out}}{2} \right] + \frac{\frac{V_{out}}{2} - V_{out}}{12\Omega}$$

$$2V_{in} - V_{out} = \frac{1}{2} \frac{dV_{out}}{dt} + \frac{1}{2} V_{out} - V_{out}$$

$$4V_{in} = \frac{dV_{out}}{dt} + V_{out}$$

7.8



KCL at non-inverting input:

$$\frac{V_{in} - V_c}{10\Omega} = (2F) \frac{dV_c}{dt}$$

Op-amp rules:

voltage at inverting input = voltage at non-inverting input = V_c

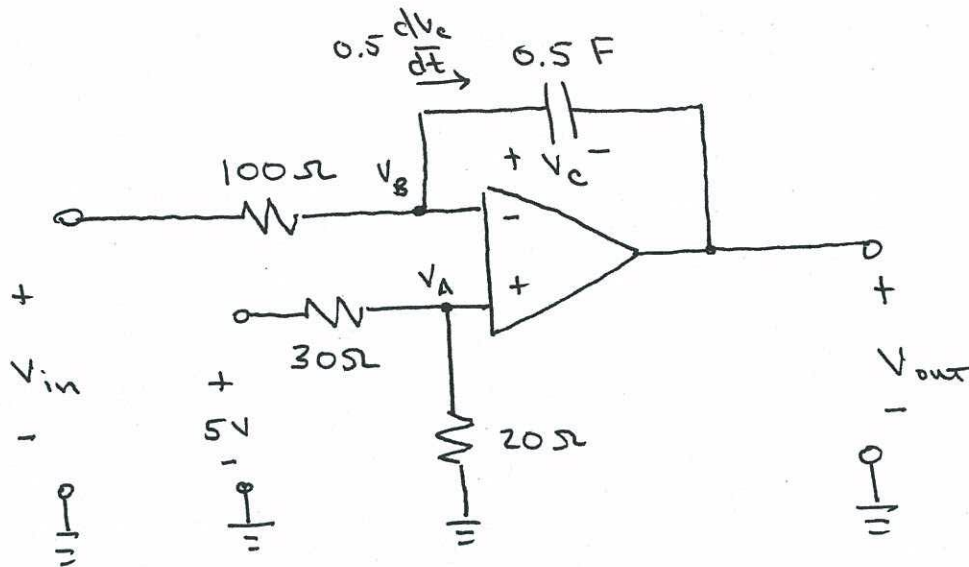
KCL at inverting input:

$$\frac{0 - V_c}{100\Omega} = \frac{V_c - V_{out}}{100\Omega} \Rightarrow V_{out} = 2V_c \Rightarrow V_c = \frac{V_{out}}{2}$$

$$\frac{V_{in} - \frac{V_{out}}{2}}{10\Omega} = (2F) \left[\frac{1}{2} \frac{dV_{out}}{dt} \right]$$

$$\boxed{\frac{V_{in}}{10} = \frac{V_{out}}{20} + \frac{dV_{out}}{dt}}$$

7.9



KCL at non-inverting input:

$$V_A = \frac{2}{5}(5\text{ V}) = 2\text{ V} \quad (\text{voltage divider})$$

Op-amp rules: $V_A = V_B = 2\text{ V}$

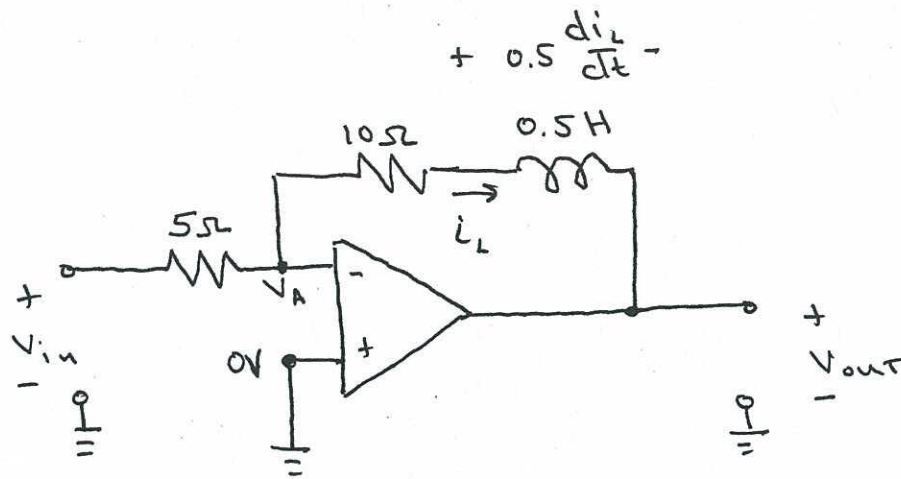
KCL at inverting input

$$\frac{V_{in} - V_B}{100\Omega} = 0.5 \frac{dV_c}{dt} = 0.5 \frac{d}{dt} [\overset{2\text{ V}}{\cancel{V_B}} - V_{out}]$$

$$V_{in} = V_B - 50 \frac{dV_{out}}{dt}$$

$$V_{in} = 2 - 50 \frac{dV_{out}}{dt}$$

7.10



Op-amp rules: $V_A = 0V$

KCL at inverting input:

$$\frac{V_{in} - V_A}{5\Omega} = i_L \Rightarrow i_L = \frac{V_{in}}{5}$$

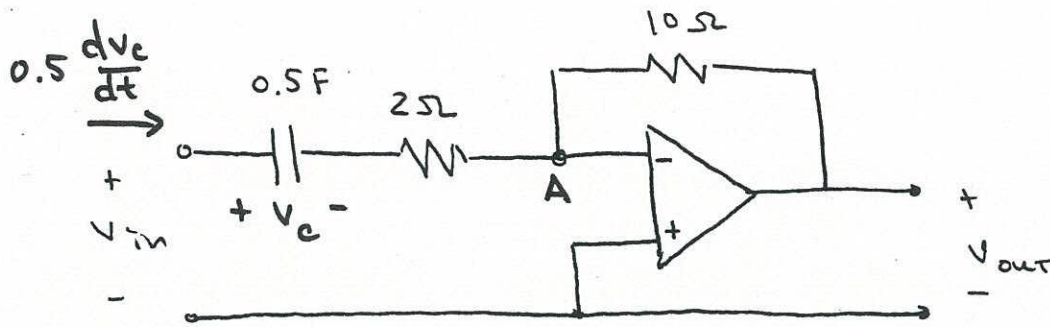
KVL, feedback loop:

$$\underbrace{V_A}_{0V} + (10\Omega)i_L + (0.5H)\frac{di_L}{dt} = -V_{out}$$

$$(10\Omega)\left(\frac{V_{in}}{5}\right) + (0.5H)\left[\frac{1}{5}\frac{dV_{in}}{dt}\right] = -V_{out}$$

$$\boxed{2V_{in} + 0.1\frac{dV_{in}}{dt} = -V_{out}}$$

7.11



Op-amp rules: $V_A = 0V$

KCL at node "A":

$$0.5 \frac{dV_c}{dt} = \frac{0V - V_{out}}{10\Omega} \Rightarrow 5 \frac{dV_c}{dt} = -V_{out} \Rightarrow \frac{dV_c}{dt} = -\frac{1}{5} V_{out}$$

KVL:

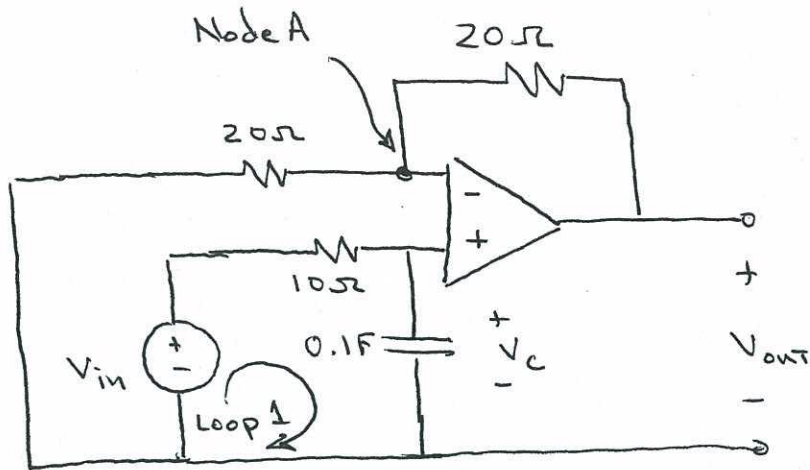
$$V_{in} = V_c + (2\Omega) \left(0.5 \frac{dV_c}{dt} \right) \Rightarrow V_{in} = V_c + \frac{dV_c}{dt}$$

$$\frac{dV_{in}}{dt} = \frac{dV_c}{dt} + \frac{d^2 V_c}{dt^2}$$

$$\frac{dV_{in}}{dt} = \left(-\frac{1}{5} V_{out} \right) + \left(-\frac{1}{5} \frac{dV_{out}}{dt} \right)$$

$$\boxed{\frac{dV_{in}}{dt} = -\frac{1}{5} \left[V_{out} + \frac{dV_{out}}{dt} \right]}$$

7.12



Voltage at Node A is V_c (from op-amp rules)

KCL at A:

$$\frac{V_c - 0}{20\Omega} + \frac{V_c - V_{OUT}}{20\Omega} = 0 \Rightarrow 2V_c = V_{OUT} \Rightarrow V_c = \frac{V_{OUT}}{2}$$

KVL, Loop 1:

$$V_{in} = 10\Omega \left[(0.1F) \frac{dV_c}{dt} \right] + V_c \Rightarrow V_{in} = \frac{dV_c}{dt} + V_c$$

Combine:

$$\frac{dV_{OUT}}{dt} + V_{OUT} = 2V_{in}$$

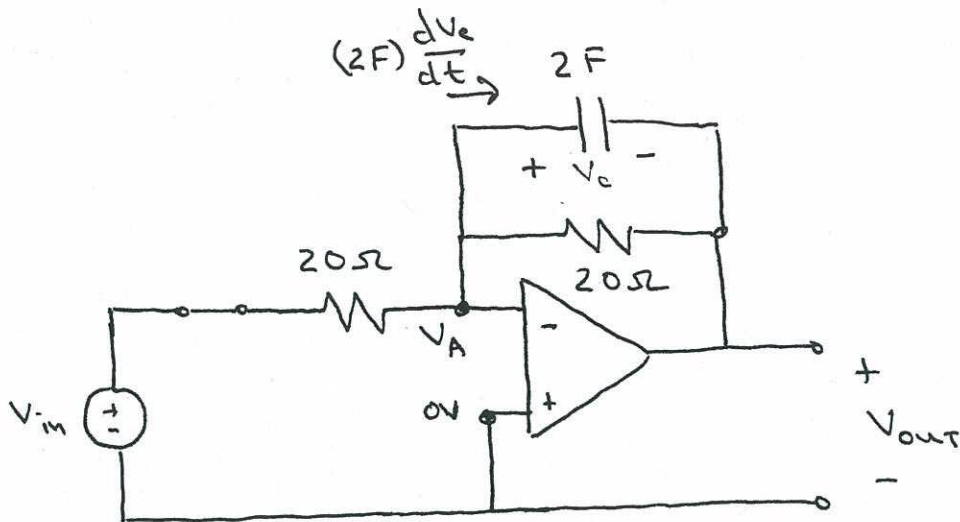
$$\tau = 1 ; V_{OUT}(t \rightarrow \infty) = 2V_{in} \Rightarrow$$

$$t=0: V_{OUT} = 0$$

$$\underline{\underline{V_{OUT} = 2V_{in} [1 - e^{-t}]}}$$

7.13

(a)

Op-amp rules : $V_A = 0V$

KCL at "A" :

$$\left. \begin{aligned} \frac{V_{in} - 0V}{20\Omega} &= \frac{0 - V_{out}}{20\Omega} + (2F) \frac{dV_c}{dt} \\ V_c &= 0 - V_{out} \end{aligned} \right\} \Rightarrow \boxed{V_{in} = -V_{out} - 40 \frac{dV_{out}}{dt}}$$

(b) Governing Diff. eqn. is first order with constant forcing function. Solution is of the form:

$$V_{out}(t) = K_1 + K_2 e^{-t/\tau}$$

Equation from (a) can be written: $-\frac{1}{40} V_{in} = \frac{1}{40} V_{out} + \frac{dV_{out}}{dt}$

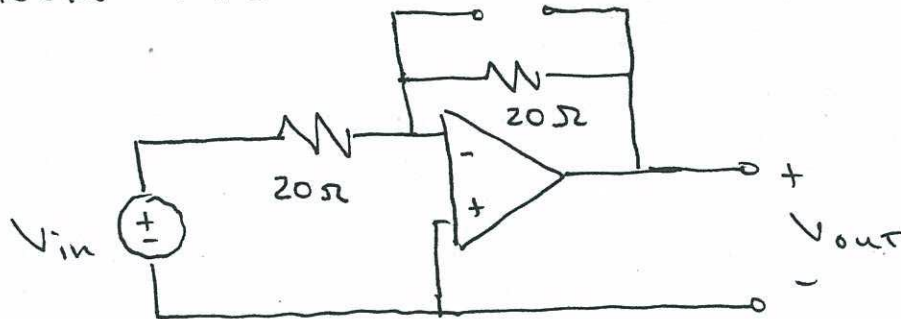
$$\Rightarrow \tau = 40 \text{ seconds}$$

Initial condition : $V_{out}(t=0) = 0V$
 Final condition : $V_{out}(t \rightarrow \infty) = -V_{in}$

$$\boxed{V_{out}(t) = -V_{in} [1 - e^{-t/40}]}$$

7.13 (cont'd)

(c) For constant forcing function, as $t \rightarrow \infty$ the capacitor behaves like an open-circuit. Thus, as $t \rightarrow \infty$, the circuit looks like:



This is an inverting voltage amplifier with gain = 1. Thus,

$$V_{out} = -V_{in}$$

From part (b), as $t \rightarrow \infty$:

$$V_{out}(\infty) = -V_{in} [1 - e^{-\infty/40}] = -V_{in}$$

They agree!